Misc

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1.0.1 "Which way standard form"

$$X \leq Y \Rightarrow OBJECTIVE + \lambda(Y - X)$$

1.0.2 "Utility Functions"

- Log
  - Doesn't have any relationship with the interest rate
- CRRA
  - $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$
  - Note: $\lim_{\theta \to 1} \frac{c^{1-\theta}-1}{1-\theta} = \ln c$
- CARA?

1.0.3 "Intertemporal elasticity of substitution"

$\frac{1}{\theta} = IES$.

- High $\theta \Rightarrow$ really want to smooth consumption. Won't be very reactive to rate changes.
- Low $\theta \Rightarrow$ don't care about smooth consumption. So, very reactive to changes in r.

1.0.4 "Production Functions"

- Cobb-Douglas: $Y_t = K_t^{\alpha} L_t^{1-\alpha}$
- Linear: $Y = AK_t + BL_t$
  - CRS, but NOT neoclassical (because Marginal products are positive, but not falling!)
  - Production-side equilibrium:
    - $R_t = A; w_t = B$

1.0.5 "Caution: log-differentiating works only in continuous time!"

In discrete time you need to use the definition of growth rates...

1.0.6 "Marginal Utility"

If $MU(c)$ is high then c is low! This makes sense if you plot it out with MU(c) on the y-axis and c on the x-axis and note that MU is falling in c.

1.0.7 "Expectations and Logs"

$$\ln(E[XY]) \neq \ln E[\ln(X) + \ln(Y)]$$

Recall Jensen's inequality:

$$\ln(\mathrm{E}[X]) \leq \mathrm{E}\left[\ln(X)\right]$$

1.0.8 "Induction"

1. Base Case: show that $p(n)$ is true for the smallest possible value of n, n_0.
2. Induction Hypothesis: Assume that the statement $p(n)$ is true for any positive integer $n = k$, for $k \geq n_0$.
3. Inductive Step: Show that the statement $p(n)$ is true for $n = k + 1$.

1.0.9 "Standard form for maximization"

Given constraint:

$$LHS \leq RHS$$

Write Lagrangian as

$$\mathcal{L} = OBJ + \lambda[RHS - LHS]$$

1.0.10 "KKT Algorithm 2 Constraints"

Given want to max

$$\mathcal{L} = f(x, y, z) + \lambda_1(constraint) + \lambda_2(constraint)$$

Step 0:

- Is the function strictly concave? If so, you will automatically have a single interior solution – assume all nonnegativity constraints are zero and solve normally.... I THINK...

Step 1:

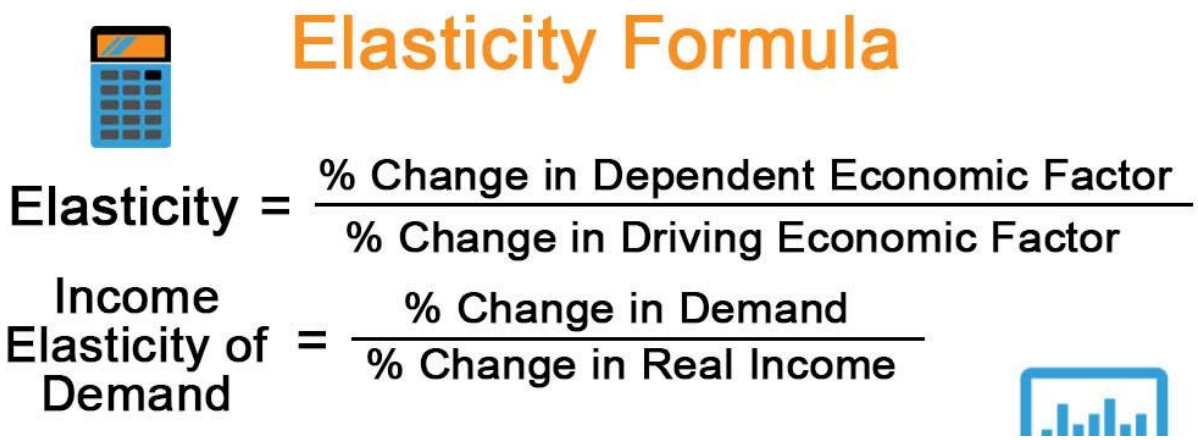
- assume constraint 2 slack ($\lambda_2 = 0$)
- solve
- test if constraint 2 really was satisfied
  - if so, stop here
  - if not, continue

Step 2:

- assume constraint 1 slack ($\lambda_1 = 0$)
- solve
- test if constraint 1 really was satisfied
  - if so, stop here
  - if not, continue

Step 3:

- Solve with both constraints non-binding (i.e., $\lambda_1 = \lambda_2 = 0$)

  "$-$" in front of objective if minimizing

1.0.11 "Elasticity"